Exp 26 MATLAB CODE

```
clc;
clear;
close all;

% Generator Matrix for Hamming (7,4) Code
G = [1 0 0 0 1 1 0;
    0 1 0 0 1 0 1;
    0 0 1 0 0 1 1;
    0 0 0 1 1 1 1];

% Parity Check Matrix
H = [1 1 0 1 1 0 0;
    1 0 1 1 0 1 0;
    0 1 1 1 0 0 1];

% Input Message
msg = [1 0 1 1];

% Encoding
codeword = mod(msg * G, 2);

disp('Original Message:');
disp(msg);

disp('Encoded Codeword:');
disp(codeword);
```

```
% Introduce Single-Bit Error
received = codeword;
error_position = 3;
received(error_position) = 1 - received(error_position);

disp('Received Codeword with Error:');
disp(received);

% Calculate Syndrome
syndrome = mod(received * H', 2);

disp('Syndrome Vector:');
disp(syndrome);

% Error Detection and Correction
error_position = 0;

for i = 1:7
    if isequal(syndrome, H(:,i)')
        error_position = i;
        break;
    end
end

if error_position ~= 0
    received(error_position) = 1 - received(error_position);
    disp('Single-Bit Error Detected and Corrected.');
else
```

```
    disp('No Error Detected.');
end

disp('Corrected Codeword:');
disp(received);

disp('Recovered Message:');
disp(received(1:4));

% Verify Correction
if isequal(received, codeword)
    disp('Error Correction Successful.');
else
    disp('Error Correction Failed.');
end
```

OUTPUT

```
Original Message:
     1     0     1     1

Encoded Codeword:
     1     0     1     1     0     1     0

Received Codeword with Error:
     1     0     0     1     0     1     0

Syndrome Vector:
     0     1     1


Single-Bit Error Detected and Corrected.
Corrected Codeword:
     1     0     1     1     0     1     0
```

Recovered Message:

1 0 1 1

Error Correction Successful.

>>